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How do income-driven repayment plans benefit student debt borrowers?

Contents

1 Big picture

- **Question.** Why did student debt rise so much in the United States, what role did income-driven repayment (IDR) plans play in that rise, and how should repayment rules be designed if the objective is to improve borrower welfare efficiently?
- **Setting.** The paper combines administrative credit-bureau evidence on repayment behavior with a life-cycle model of borrowers who face incomplete markets, idiosyncratic income risk, and institutional student-loan repayment rules.
- **Core challenge.** Lower payments today can help borrowers by relaxing liquidity constraints and insuring downside income risk, but they can also simply transfer resources from taxpayers to borrowers. A good policy evaluation has to separate these channels.
- **Main positive result.** Deferred repayment explains a very large share of the rise in student debt balances. The paper estimates that slower repayment accounts for roughly **$391 billion**, or **44%**, of the increase in student debt balances between 2010 and 2020.
- **Main welfare result.** Existing IDR rules generate large borrower welfare gains—about **$50,600** per borrower on average—but about half of that gain comes from taxpayer subsidization rather than pure efficiency gains from smoothing and insurance.
- **Main normative result.** Current IDR rules are not cost-efficient. A much longer repayment horizon paired with a more progressive repayment schedule can deliver similar or higher borrower welfare at dramatically lower taxpayer cost. The net-welfare-maximizing policy reduces taxpayer cost by about **$19,700** per borrower relative to the status quo while leaving borrowers roughly indifferent and even slightly better off.
- **Policy takeaway.** The paper argues that if the government wants to subsidize education, it should not do most of that subsidization through very generous IDR rules. IDR should primarily be used to provide **insurance and intertemporal smoothing**, not large transfers.
- **Why this paper matters.** It reframes student debt as a **repayment-design problem**, not just a borrowing-amount problem. The paper also provides a general welfare-decomposition method that can be applied to other public insurance and repayment programs.

2 Data and measurement (key objects)

2.1 Observed borrower-level administrative panel

- **Main data source.** The paper uses the **Booth TransUnion Consumer Credit Panel**, an anonymized 10% sample of credit records from 2000 to 2020.
- **Unit of observation.** Individual borrower i over time t.
- **Observed loan objects.** The data include original balance, current balance, scheduled payments, loan maturity, and the date at which a borrower first enters repayment.
- **Repayment cohorts.** Borrowers are assigned to repayment cohorts by the first year they are scheduled to begin repayment.

- **Aggregate moments.** The paper constructs aggregate student debt balances, numbers of borrowers, average balances, repayment profiles by cohort, and counterfactual balances under the standard ten-year plan.

2.2 Institutional repayment environment

- **Standard plan.** In the benchmark institutional environment, borrowers repay through fixed payments over ten years.
- **Default / garnishment.** When borrowers do not repay under the standard plan, the government can garnish a portion of earnings, subject to legal caps.
- **IDR expansion.** Beginning in 2008, the US introduced much more generous IDR plans. The paper treats the evolution from standard repayment to IBR, PAYE, REPAYE, SAVE, and RAP as central institutional changes.
- **SAVE.** SAVE raised the repayment threshold to 225% of the federal poverty line, reduced undergraduate repayment rates to 5%, shortened time to forgiveness for small undergraduate balances, and eliminated negative amortization.
- **RAP.** RAP extended repayment before forgiveness to 30 years and replaced the standard IDR formula with a graduated schedule tied directly to income, plus a child subsidy.

2.3 State variables and calibration objects in the model

- **Financial state variables.** Borrowers are characterized by gross financial wealth W_{it}, remaining loan balance L_{it}, and a persistent earnings component z_{it}.
- **Initial condition.** Borrowers begin adulthood with debt at graduation L_{i,t_0}; the paper discretizes initial debt into eleven groups.
- **Income.** Labor earnings are the product of an aggregate component and an idiosyncratic component with persistent, transitory, and unemployment shocks.
- **Household composition.** Household size and dependents matter because the poverty line, SNAP eligibility, and the repayment threshold all depend on family composition.
- **Calibration inputs.** The model is calibrated using TransUnion loan data, the SCF for wealth and household structure, and Guvenen et al. style earnings dynamics for labor-income risk.

2.4 Empirical applications

- Decomposition of the rise in balances into borrowing versus deferred repayment.
- Welfare decomposition of existing IDR rules.
- Search over optimal policy parameters under budget-neutral and net-welfare criteria.
- Evaluation of recent policy proposals, especially SAVE and RAP.
- Discussion of moral hazard in borrowing and small labor-supply responses.

3 Models and empirical specifications (all equations + interpretation)

3.1 Counterfactual debt accounting under the standard plan

To isolate the contribution of slower repayment to the rise in aggregate balances, the paper constructs a borrower-level counterfactual balance path under standard-plan amortization:

$$L_{i,t+1}^{CF} = L_{it}^{CF}(1 + r_{L,i}) - R_{it}^{std},$$

where the standard annual payment is

$$R_{it}^{std} = \frac{r_{L,i} L_{i,t_0}^{CF}}{1 - (1 + r_{L,i})^{-10}}.$$

- **Interpretation.** The counterfactual fixes each cohort's initial balance and interest rate and asks: what would balances look like if borrowers kept repaying on the old ten-year schedule?
- **Why this matters.** This calculation strips out some obvious confounds. Tuition, interest rates, and borrowing levels can vary across cohorts, but they are already built into the initial balance and rate. The remaining gap is attributable to repayment slowdown.

3.2 Benchmark economy without IDR

In the benchmark economy, balances evolve according to

$$L_{i,t+1} = L_{it}(1 + r_L) - R_{L,it}.$$

Under the standard plan, the required payment is fixed:

$$R_{it}^{std} = \frac{r_L L_{i,t_0}}{1 - (1 + r_L)^{-10}}.$$

If the borrower fails to repay, the government can garnish income:

$$R_{it}^{garnish} = \min\{0.15(Y_{it} - \Gamma_{it}),\ R_{it}^{max\, garnish}\},$$

with

$$R_{it}^{max\, garnish} = \min\{0.25(Y_{it} - \Gamma_{it}),\ \lambda_{garnish} Y_{1,t}\}.$$

- **Interpretation.** The no-IDR benchmark is not a world without student loans. It is a world with student loans, fixed repayment, and default/garnishment risk.
- **Economic role.** This benchmark gives the paper a clean comparison point for measuring how much welfare comes specifically from introducing flexible repayment.

3.3 Economy with IDR

With IDR, borrowers may repay a fraction of discretionary income up to a cap given by the standard-plan payment, and remaining debt is forgiven after a fixed number of years:

$$R_{it}^{idr} = \begin{cases} \min\left\{\theta_{idr}(Y_{it} - \lambda_{idr} Y_{pl,t})_+,\ R_{it}^{std}\right\}, & t \le t_F^{idr}, \\ 0, & t > t_F^{idr}. \end{cases}$$

The paper's baseline current-rule calibration uses

$$\lambda_{idr} = 150\% \text{ of the poverty line}, \qquad \theta_{idr} = 10\%, \qquad t_F^{idr} = 20 \text{ years}.$$

Borrowers can also prepay, and the model assumes that recent administrative simplifications make opting into IDR dominant relative to going into garnishment.

- **Interpretation.** IDR changes both the **time profile** and the **state contingency** of payments.
- **Key trade-off.** Lower payments in bad states and early years improve borrower welfare, but they reduce present-value repayments and can increase taxpayer cost.

3.4 Borrower problem

Borrowers maximize expected discounted utility:

$$V_{i,t_0} = \mathbb{E} \sum_{t=t_0}^{T} \beta^{t-t_0} \Big(\prod_{k=t_0}^{t} (1 - m_k) \Big) u(C_{it}),$$

with period utility

$$u(C_{it}) = \frac{1}{1-\gamma} \left(\frac{C_{it}}{\sqrt{N_{it}}} \right)^{1-\gamma}.$$

Gross financial wealth evolves as

$$W_{i,t+1} = \big(W_{it} - R_{L,it} + Y_{it} + B_{it}^{SS} + B_{it}^{SN} - \Gamma_{it} - C_{it} \big)(1 + r_f).$$

The recursive problem is

$$V_{it}(W_{it}, L_{it}, z_{it}; L_{i0}) = \max_{C_{it}, R_{L,it}, x_{it}} \left\{ \frac{1}{1-\gamma} \left(\frac{C_{it}}{\sqrt{N_{it}}} \right)^{1-\gamma} + \beta(1 - m_t) \mathbb{E}\big[V_{i,t+1}(W_{i,t+1}, L_{i,t+1}, z_{i,t+1}; L_{i0}) \big] \right\}.$$

subject to the law of motion for wealth and debt and the choice of repayment regime $x_{it} \in \{std, idr\}$.

- **Interpretation.** This is a standard incomplete-markets life-cycle problem augmented with institutional repayment rules.
- **Key insight.** IDR affects welfare because borrowers cannot freely borrow against future income or perfectly insure idiosyncratic shocks. Flexible repayment partly substitutes for missing financial markets.

3.5 Income, taxes, and social insurance

Labor earnings are decomposed as

$$Y_{it} = Y_{1,t} \cdot Y_{2,it},$$

where $Y_{1,t}$ is the aggregate wage index and $Y_{2,it}$ is the idiosyncratic component with deterministic age profile, persistent AR(1) component, transitory shocks, and unemployment risk.

The model also includes:

- **SNAP / food stamps,** which prevent extreme disaster-state consumption outcomes,
- **Social Security taxes and retirement benefits,** which matter for the lifetime budget and for how long debt can realistically be repaid,
- **Progressive income taxation,** which interacts with repayment rules when computing net resources and fiscal cost.

- **Interpretation.** The paper does not evaluate student loans in isolation. It embeds them inside a broader public-insurance system.
- **Why it matters.** Some welfare gains from IDR would look mechanically huge if the model ignored other safety-net programs. Including them makes the exercise more realistic.

3.6 First-best benchmark and sources of welfare loss

In frictionless complete markets, the optimal consumption plan equalizes discounted marginal utility across states and periods:

$$\beta^{t-t_0}\Big(\prod_{k=t_0}^{t}(1-m_k)\Big)\frac{u'(C^*_{its})}{P_{t_0,t}} = u'(C^*_{i,t_0}).$$

With incomplete markets, the welfare loss relative to the complete-market benchmark can be approximated by

$$V^*_{i,t_0} - V_{i,t_0} \approx \sum_t \mathbb{E}\big[v'_{it}(C^*_{it} - C_{it})P_{t_0,t}\big],$$

which the paper decomposes into:

- **imperfect insurance** across states of the world,
- **imperfect intertemporal smoothing** over the life cycle,
- and, at the aggregate level, **inequality** across borrowers.

A useful representation is

$$V^* - V \approx \sum_t \mathrm{Cov}_i\big(\mathbb{E}[v'_i], \mathbb{E}[(C^*_{it} - C_{it})P_{t_0,t}]\big) + \sum_t \mathbb{E}\big[\mathrm{Cov}_{it}(v'_i, (C^*_{it} - C_{it})P_{t_0,t})\big].$$

- **Interpretation.** Welfare gains do not arise because policymakers lower debt per se; they arise when policy moves consumption toward states, dates, or borrowers with high marginal utility.
- **Main conceptual contribution.** This section is the backbone of the paper. It separates liquidity/insurance gains from pure redistribution.

3.7 Policy welfare decomposition

For a policy p, borrower welfare gains are approximated by

$$\Delta_p V \approx \sum_i \sum_t \mathbb{E}\big[v'\Delta C\, P_{t_0,t}\big].$$

Net welfare subtracts the taxpayer cost scaled by the marginal cost of public funds (MCPF):

$$\Delta_p^{net} V = \Delta_p V - \text{MCPF} \cdot \sum_t -\mathbb{E}[v'_\Gamma]\, \Delta R_{L,t} P_{t_0,t}.$$

The paper then decomposes net welfare into five components:

$$\begin{aligned}\Delta_p^{net} V = &\underbrace{\text{insurance}}_{\text{across states}} + \underbrace{\text{intertemporal smoothing}}_{\text{across time}} \\ &+ \underbrace{\text{transfer progressivity}}_{\text{across borrowers}} + \underbrace{\text{mean transfer}}_{\text{taxpayers to borrowers}} \\ &- \underbrace{\text{fiscal externality}}_{(\text{MCPF}-1)\times\text{transfers}}.\end{aligned}$$

In covariance form, the paper writes

$$\begin{aligned}\Delta_p^{net} V = &\frac{1}{I}\sum_i \sum_t \text{Cov}_{it}(v', \Delta CP_{t_0,t}) \\ &+ \frac{T}{I}\sum_i \text{Cov}_i\big(\mathbb{E}[v'], \mathbb{E}[\Delta CP_{t_0,t}]\big) \\ &+ T\text{Cov}\big(\mathbb{E}[v'], \mathbb{E}[\Delta CP_{t_0,t}]\big) \\ &- (\text{MCPF} - 1)\sum_t -\mathbb{E}[v']\Delta R_{L,t} P_{t_0,t}.\end{aligned}$$

- **Interpretation.** This decomposition is what lets the paper say whether a reform is genuinely efficient or merely generous.
- **Operational advantage.** The decomposition can be implemented using only the baseline simulation and the policy counterfactual; it does not require solving a new full planner problem for each component.

3.8 Baseline calibration and welfare under current IDR rules

The paper calibrates preferences using Gourinchas and Parker style values, with $\beta = 0.96$ and risk aversion $\gamma = 2$. The real risk-free rate is set to $r_f = 0.02$, while student-loan interest rates vary with debt level to mimic undergraduate, graduate, and PLUS borrowing.

Under current IDR rules, the decomposition of borrower welfare gains is:

- total borrower welfare gain: 0.793,
- insurance: 0.169,
- intertemporal smoothing: 0.237,
- progressivity: 0.002,
- mean transfer from taxpayers: 0.377.

Using the 2022 national wage index, this maps into about $50,600 per borrower.

- **Interpretation.** Roughly half of current-IDR welfare gains come from taxpayer transfers and the other half from genuine efficiency gains.
- **Important nuance.** The program is almost neither progressive nor regressive in aggregate because high-debt borrowers tend also to be high earners.

3.9 Comparative statics of IDR parameters

The three central policy parameters are:

- repayment years before forgiveness t_F^{idr},
- repayment threshold λ_{idr},
- repayment rate on discretionary income θ_{idr}.

The paper studies the welfare derivatives of these parameters relative to fiscal cost.

The main lessons are:

- extending repayment years lowers taxpayer transfers but does *not* reduce the intertemporal-smoothing value of IDR,
- raising the payment threshold is the most cost-efficient way to increase borrower welfare,
- raising the payment rate can actually improve insurance and progressivity while containing taxpayer cost,
- cutting time to forgiveness or lowering interest rates is a relatively inefficient way to spend taxpayer dollars.

- **Interpretation.** The core policy insight is that generosity should come through **better timing and targeting of payments**, not mostly through earlier forgiveness.

3.10 Optimal policy design

The paper searches over $\{t_F^{idr}, \lambda_{idr}, \theta_{idr}\}$, holding student-loan interest rates fixed. It studies two policies:

- a **budget-neutral** policy,
- and a policy that **maximizes borrower welfare net of taxpayer cost and fiscal externalities**.

The budget-neutral optimum is:

$$t_F^{idr} = 43, \qquad \lambda_{idr} = 131\%, \qquad \theta_{idr} = 23.6\%.$$

The highest-net-welfare optimum is:

$$t_F^{idr} = 43, \qquad \lambda_{idr} = 237\%, \qquad \theta_{idr} = 24.4\%.$$

The corresponding welfare decompositions are:

$$\text{Budget-neutral: } (0.604,\ 0.307,\ 0.274,\ 0.023,\ 0.000),$$

$$\text{Highest net welfare: } (0.808,\ 0.329,\ 0.374,\ 0.037,\ 0.068),$$

where each tuple is {total borrower welfare, insurance, intertemporal smoothing, progressivity, mean transfer}.

- **Interpretation.** The paper's preferred policy is much less subsidized than the status quo, but more progressive in timing. Borrowers pay for longer, but only once they are in better states.
- **Efficiency logic.** A dollar saved when young or in a low-income state is worth much more to the borrower than a dollar repaid later in life when income is high. This is why long maturities and high thresholds are so powerful.

3.11 SAVE and RAP in the model

SAVE modifies the repayment rule to

$$R_{it}^{save} = \begin{cases} \theta_{save,i}(Y_{it} - \lambda_{save} Y_{pl,t})_+, & t \le t_{F,i}^{save}, \\ 0, & t > t_{F,i}^{save}, \end{cases}$$

with no standard-plan cap and no negative amortization:

$$L_{i,t+1} = \min\{L_{it}(1 + r_L) - R_{it}^{save},\ L_{it}\}.$$

RAP implies

$$R_{it}^{rap} = \begin{cases} \theta_{rap,i}(Y_{it})\, Y_{it} - \delta_{kids} n_{kids}, & t \le t_F^{rap}, \\ 0, & t > t_F^{rap}, \end{cases}$$

with balances evolving as

$$L_{i,t+1} = \min\{L_{it}(1 + r_L) - R_{it}^{rap},\ L_{it} - \delta\}.$$

- **Interpretation.** SAVE mainly pushes the system toward more transfer-heavy generosity. RAP moves in the right direction by extending maturity, but not far enough to reach the paper's welfare frontier.

3.12 Borrowing incentives and labor-supply discussion

The paper studies the cost of debt under different rules and defines a lifetime marginal tax rate (LMTR):

$$LMTR_i \equiv \frac{\sum_t (1 + r)^{-t} \mathbb{E}_t[\Delta \Gamma_{i,t}]}{\sum_t (1 + r)^{-t} \mathbb{E}_t[\Delta Y_{i,t}]},$$

and maps this into labor-supply changes with elasticity ε:

$$\frac{\Delta h_{i,p}}{h_{i,0}} = \left(\frac{1 - LMTR_{i,p}}{1 - LMTR_{i,0}} \right)^{\varepsilon} - 1.$$

- **Interpretation.** This is a robustness exercise, not the main engine of the paper.
- **Result.** Labor-supply responses are tiny. SAVE slightly lowers labor supply, while the optimal policies slightly increase it because they reduce the probability that debt is ultimately forgiven and hence reduce moral hazard.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in a naive interpretation

- It would be incorrect to interpret the rise in student debt balances as purely a borrowing shock. Part of the rise comes from **slower amortization** rather than larger original loans.
- It would also be incorrect to interpret lower required payments as mechanically welfare improving. Some of those lower payments are just taxpayer transfers, while others are true improvements in smoothing and insurance.
- Likewise, it would be incorrect to assume that more generous rules are automatically better. The paper shows that some generosity margins are very expensive and not very efficient.

4.2 Why the deferred-payment decomposition is informative

- The counterfactual balance path holds fixed initial debt, cohort, and interest-rate conditions, and changes only the repayment rule.
- The divergence between actual and counterfactual balances begins around the institutional rollout of IDR plans and accelerates around PAYE and REPAYE, which strengthens the interpretation that repayment reforms matter.
- Because the comparison is done cohort by cohort, it is not simply capturing aggregate enrollment or tuition trends.

4.3 Why the welfare decomposition is informative

- The model builds a complete-market benchmark and compares actual consumption allocations to that benchmark.
- The decomposition traces welfare gains to covariances of consumption changes with marginal utility across states, dates, and borrowers. That is exactly the right metric for a CRRA life-cycle environment.
- The method distinguishes four channels cleanly: insurance, intertemporal smoothing, transfer progressivity, and average transfers from taxpayers.

4.4 Why the optimal-policy exercise is informative

- The policy design problem is well defined because the paper restricts attention to the key IDR parameters actually under the control of repayment policy: years before forgiveness, repayment threshold, and repayment rate.
- It imposes realistic institutional constraints, including a maximum repayment horizon up to retirement and an upper bound on repayment rates so total effective tax rates do not exceed 100%.
- The auxiliary-model approach lets the authors map the welfare surface over policy parameters without repeatedly solving the full structural model at every grid point.

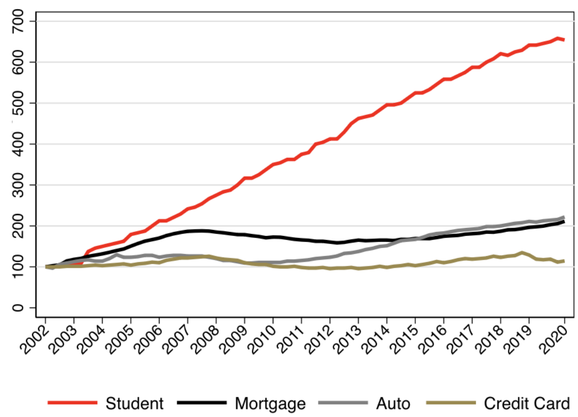

Fig. 1. Household debt in the United States 2004–2020.
This figure displays the growth in mortgage, auto loan, student loan, and credit card balances for consumers from 2002 to 2020. Balances are reported from the end of each fiscal quarter. Aggregate consumer balances are normalized to 100 in 2002.
Source: Data from the Federal Reserve Bank of New York.

4.5 Main limitations and what they imply

- **Exogenous debt at graduation.** The model does not let students re-optimize borrowing when IDR rules change. This matters especially for SAVE, which may induce much stronger borrowing responses.
- **Small labor-supply effects rather than fully endogenous labor supply.** The paper argues these effects are likely small in the US institutional setting, and the robustness exercise confirms that they are quantitatively minor.
- **Steady-state focus.** The analysis abstracts from transition-era frictions, borrower confusion, and policy uncertainty, which were important in practice during the rollout of real-world IDR plans.

4.6 Relation to the literature

- The paper connects to the literature on student-loan repayment plans, forgiveness, and forbearance.
- It also speaks to structural work on student debt and human-capital finance.
- More broadly, it fits into the literature on public insurance and liquidity provision in incomplete-markets life-cycle models.

5 Tables and figures

5.1 Figure 1: Household debt in the United States

Read:

- Student debt grew much faster than mortgage, auto, or credit-card debt.

- Relative to 2002, student debt increased more than sixfold, whereas other major household-debt categories increased by less than 200%.
- This figure motivates the paper's central factual question: why did student debt behave so differently?

5.2 Figures 2 and 3: Student-loan aggregates, repayment slowdown, and negative amortization

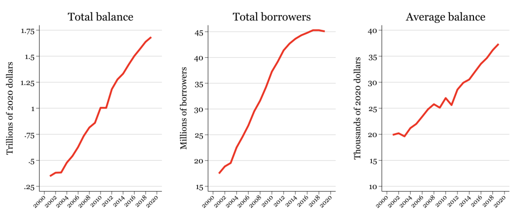

Fig. 2. Student loan aggregates.
This figure displays the aggregate student loan balance in the United States, the total number of borrowers by year, and the average balance by year, from 2001 to 2020. Numbers are as of December of each year.
Source: Data from TransUnion.

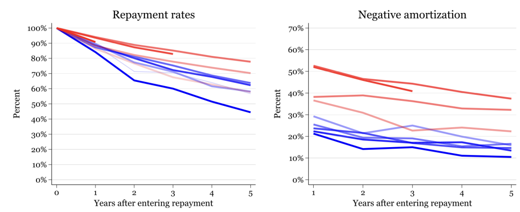

Fig. 3. Repayment rates and Negative amortization.
This figure shows the repayment behavior of every other repayment cohort from 2001 (blue) to 2019 (red). The left-hand panel indicates the repayment rate, or what percentage of the total balance owed by borrowers when entering repayment remains. The right-hand panel displays negative amortization, or the percentage of people who have an equal or higher balance than they did when entering repayment. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)
Source: Data from TransUnion.

Read:

- The rise in student debt is driven by both more borrowers and higher average balances.
- More importantly for the paper, repayment rates slowed sharply across cohorts.
- Earlier cohorts repaid close to what a ten-year plan predicts, while later cohorts kept much larger fractions of their balances outstanding.
- Negative amortization became widespread for later cohorts, meaning many borrowers' balances rose rather than fell.

5.3 Figures 4 and 5: Actual versus counterfactual balances and deferred payments by cohort

Read:

- Actual and counterfactual aggregate balances move together until about 2008, then diverge sharply.
- By 2020, actual balances are about $1.7 trillion versus a counterfactual of about $1.31 trillion.
- The implied gap is **$391 billion**, which the paper interprets as the contribution of deferred repayment.
- The cohort decomposition lines up closely with IBR, PAYE, and REPAYE rollouts, which is strong institutional evidence for the repayment-channel story.

5.4 Table 1 and Figure 7: Welfare gains under existing IDR rules and who gets them

Read:

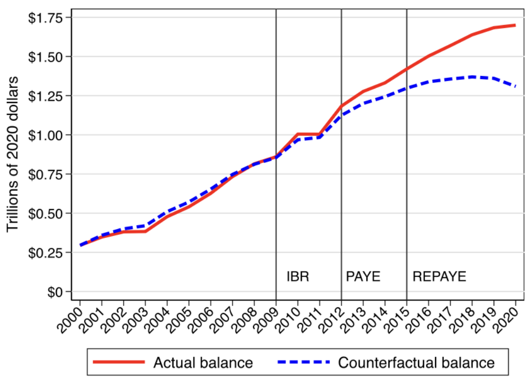

Fig. 4. Actual and counterfactual balance.
This figure displays the actual aggregate student loan balance and the counterfactual balance that would result if all students paid their loans down according to the 10-year standard plan. Vertical lines show the introduction of IDR plans.
Source: Data from TransUnion.

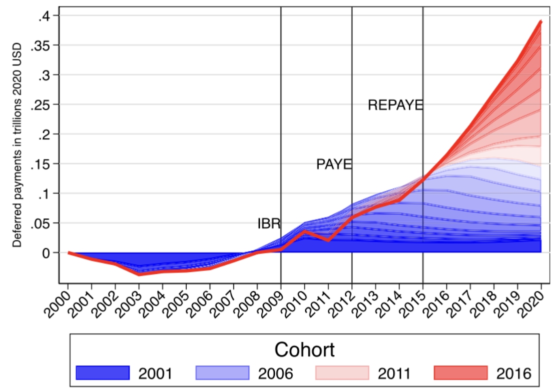

Fig. 5. Deferred payments by cohort.
This figure displays cumulative deferred payments by cohort, relative to the standard ten-year repayment plan. Vertical lines show the introduction of IDR plans. The red line represents aggregate deferred payments across cohorts. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)
Source: Data from TransUnion.

- Existing IDR raises borrower welfare by about **0.793** times the national wage index, or roughly **$50,600** per borrower.
- About **47.5%** of this gain comes from taxpayer transfers, **29.9%** from intertemporal smoothing, and **21.3%** from insurance.
- Progressivity across borrowers is basically zero in aggregate.
- Welfare gains are especially large for borrowers with high balances, but many of these borrowers also have relatively high expected lifetime earnings. This is why the program is not strongly progressive overall.

Table 1
Welfare gains to borrowers from IDR.

Total change	Financial frictions		Transfer	
in borrower welfare	Insurance	Intertemporal smoothing	Progressivity	Mean
0.793	0.169	0.237	0.002	0.377

This table reports the decomposition of aggregate welfare gains for borrowers under existing rules. The decomposition is defined in Eq. (28). Welfare gains are converted in dollar equivalent per borrower using Eq. (25), and normalized by the national wage index ($63,795 in 2022).

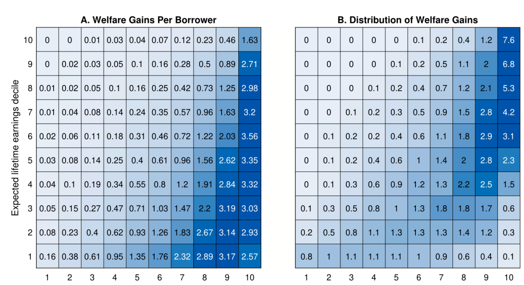

Fig. 7. Welfare Gains Per Borrower by decile of debt and income.
Panel A reports the average welfare gain from IDR per borrower as a function of their decile of expected lifetime earnings and decile of student loan at graduation. Gains are reported in multiples of the national average wage index ($63,795 in 2022). Panel B reports the share of total welfare gains from IDR by decile of expected lifetime earnings and decile of student loan at graduation, taking into account the population distribution, and in percentage of total aggregate gains for borrowers.

5.5 Figure 8 and Table 2: Which policy margins are efficient?

Read:

- Extending repayment horizons mainly cuts taxpayer transfers and does not destroy intertemporal-smoothing gains.
- Raising the repayment threshold is the most cost-efficient way to improve borrower welfare. In the normalized Jacobian, it delivers the largest total welfare gain per fiscal dollar.

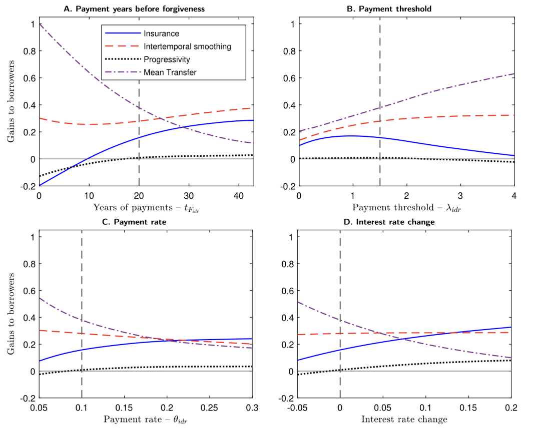

Fig. 8. Role of IDR parameters.
This figure illustrates how the four components of welfare gains evolve with four IDR parameters. Each parameter is varied independently, with the remaining three held constant at their baseline values under existing IDR rules, indicated by vertical dashed lines. Panel A reports how the four component of welfare depend on the number of payment years before forgiveness. That is the parameter $t_{F_{idr}}$ in Eq. (9). Panel B illustrates the relationship with the payment threshold, expressed as a percentage of the federal poverty line. That is the parameter λ_{idr} in Eq. (9). Panel C examines the effects of the discretionary income share allocated to payments. That is the parameter θ_{idr} in Eq. (9). Panel D explores the changes in welfare components in response to a uniform increase in the IDR loan interest rate for all loan amounts.

Table 2
Jacobian matrix of welfare components with respect to policy parameters, normalized by fiscal cost.

	Financial frictions		Transfer		Total change
	Insurance	Intertemporal smoothing	Progressivity	Mean	in borrower welfare
Years of payment ($t_{F_{idr}}$)	−0.405	−0.139	−0.085	0.714	0.086
Payment threshold (λ_{idr})	−0.246	0.299	−0.016	0.714	0.751
Payment rate (θ_{idr})	−0.345	0.145	−0.135	0.714	0.379
Loan interest rate	−0.408	−0.032	−0.187	0.714	0.087

This table reports the first-order derivative of the components of borrower welfare gains with respect to the IDR policy parameters, at the current IDR policy rule, and divided by the derivative of the fiscal cost. The components of borrower welfare gains come from the decomposition defined in Eq. (28). Welfare gains are estimated using the auxiliary model and reported per dollar cost to the government, which is equal to the change in the present value of loan payment multiplied by the MCPF of 1.4. Derivatives are based on 1 year increase in IDR years of repayment ($t_{F_{idr}}$), 10 percentage point increase in the payment threshold (λ_{idr}), 1 percentage point increase in the payment rate (θ_{idr}), and 0.1 percentage point increase in the IDR loan interest rate, in rows 1–4, respectively.

- Increasing the repayment rate can actually improve insurance and progressivity by concentrating relief on low-income states.
- Lowering interest rates or shortening time to forgiveness is comparatively inefficient—those changes mostly increase subsidies rather than efficiency.

5.6 Tables 3 and 4: Optimal IDR parameters and welfare decomposition

Table 3
Optimal IDR parameters.

Policy	Years of payment $t_{F_{idr}}$	Payment threshold λ_{idr}	Payment rate θ_{idr}
Budget-neutral	43	131%	23.6%
Highest net welfare	43	237%	24.4%
Status Quo	20	150%	10%

This table reports estimated IDR policy parameters maximizing welfare under two policy constraints. The "budget-neutral" policy maximizes borrower welfare at no cost to taxpayers relative to the benchmark economy without IDR. The "highest net welfare" policy maximizes borrower welfare net of the cost to taxpayers and fiscal externalities. The Status Quo reports parameters under existing rules.

Table 4
Welfare gains from optimal IDR parameters.

	Total change	Financial frictions		Transfer	
	in borrower welfare	Insurance	Intertemporal smoothing	Progressivity	Mean
Budget-neutral	0.604	0.307	0.274	0.023	0.000
Highest net welfare	0.808	0.329	0.374	0.037	0.068
Status Quo	0.793	0.169	0.237	0.002	0.377

This table reports the decomposition of borrower welfare gains under the optimal IDR policy parameters reported in Table 3. The decomposition is defined in Eq. (28). Welfare gains are estimated using the auxiliary model and reported in dollar equivalent per borrower and as a multiple of the national wage index ($63,795 in 2022). The Status Quo reports welfare gains under current rules.

Read:

- The **budget-neutral** optimum is extremely different from current rules: 43-year repayment, 131% threshold, 23.6% repayment rate.
- The **highest-net-welfare** optimum also uses a 43-year repayment horizon, but raises the threshold much more aggressively to 237% and sets the repayment rate at 24.4%.
- Even at zero taxpayer cost, borrower welfare remains large at about **$38,500** per borrower.
- The net-welfare-maximizing policy yields about **$51,500** of borrower welfare (0.808 wage units) with only about **$4,300** of average transfer from taxpayers.

5.7 Figure 9 and Table 5: Welfare frontier and robustness to the marginal cost of public funds

Read:

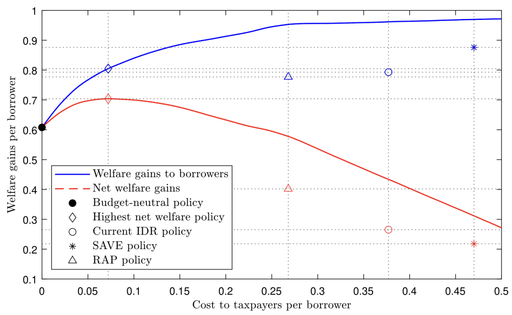

Fig. 9. Welfare Gains by policy budget under optimal calibration.
This figure illustrates the relation between the per-borrower budget allocated to IDR and welfare gains to borrowers (blue line) or net welfare gains (red dashed line) under the corresponding optimal calibration of IDR parameters. The black dot (•) marks the coordinates of the budget-neutral policy. Diamonds (◊) mark the coordinates of welfare gains under the net-welfare maximizing policy. Circles (∘), stars (∗) and triangle (△) represent current IDR rules, SAVE, and RAP rules respectively. Welfare gains and cost to taxpayers are reported per borrower and as a multiple of the national wage index ($63,795 in 2022). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 5
Marginal cost of public funds and optimal IDR policy.

MCPF	Optimal policy parameter			Optimal welfare results		
	$t_{F_{idr}}$	λ_{idr}	θ_{idr}	Total change, borrowers	Transfer, Mean	MVPF
1	43	297%	35.4%	0.850	0.104	8.2
1.2	43	246%	25.0%	0.817	0.075	9.1
1.4	43	237%	24.4%	0.808	0.068	8.4
1.6	43	226%	24.4%	0.795	0.060	8.3
1.8	43	226%	24.5%	0.795	0.059	7.5
2	43	207%	24.3%	0.767	0.045	8.5

This table reports estimated policy parameters that maximize net welfare, and the resulting optimal transfers and borrower welfare gains with varying values of the Marginal Cost of Public Funds (MCPF). The optimal policy in each row maximizes borrower welfare net of the cost to taxpayers time the MCPF, to include fiscal externalities (Eq. (29)). The value of MCPF = 1.4 resembles the baseline "Highest net welfare" results in Section 7.2 and Table 4. The last column shows the marginal value of public funds, MVPF, defined as the welfare gains to borrowers divided by direct cost to taxpayers (transfers) and the MCPF (to account for fiscal externalities).

Table 6
Welfare gains to borrowers from recent policy proposals.

	Total change	Financial frictions		Transfer	
	in borrower welfare	Insurance	Intertemporal smoothing	Progressivity	Mean
Status Quo	0.793	0.169	0.237	0.002	0.377
SAVE	0.876	0.114	0.294	−0.007	0.470
Change	0.083	−0.055	0.057	−0.009	0.093
RAP	0.777	0.216	0.276	0.016	0.268
Change	−0.016	0.047	0.039	0.014	−0.109

This table reports the decomposition of aggregate welfare gains for borrowers under existing IDR rules, SAVE and RAP. The decomposition is defined in Eq. (28). Welfare gains are converted in dollar equivalent per borrower using Eq. (25), and normalized by the national wage index ($63,795 in 2022).

- The welfare frontier shows that a large share of current-IDR borrower welfare can be delivered at little or no taxpayer cost.
- Net welfare peaks around a per-borrower subsidy of **0.068 wage units**, about **$4,300**.
- The figure makes the paper's central normative point visually: current IDR is far to the right of the efficient frontier.
- When MCPF rises from 1 to 2, the optimal repayment horizon remains 43 years, the threshold falls from 297% to 207%, and the MVPF stays around 8. So the paper's main conclusion is robust to reasonable public-finance assumptions.

5.8 Table 6: SAVE and RAP relative to the status quo

Read:

- SAVE raises borrower welfare by about **$5,300**, but that entire gain comes from bigger taxpayer transfers. Insurance gains actually fall.
- RAP reduces borrower welfare only slightly—about **$1,000**—while saving taxpayers about **$7,000** per borrower.
- RAP therefore moves in the direction favored by the model, because it extends repayment horizons and reduces pure subsidization.
- But RAP is still not fully cost-efficient relative to the optimized policies.

5.9 Figures 10 and 11: Cost of debt and borrowing incentives

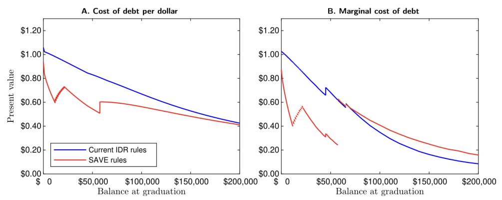

Fig. 10. Cost of student debt under existing rules and SAVE.
Panel A reports the average simulated present value of student debt at graduation, per dollar of debt, under existing IDR rules and with SAVE. Panel B reports the mean marginal cost of student debt, defined as the present value of borrowing one more dollar just before graduation. The marginal cost of debt is infinite at points where the number of repayment years increases. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

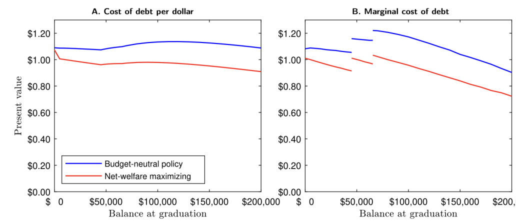

Fig. 11. Cost of student debt with optimal policy parameters.
Panel A reports the average simulated present value of student debt at graduation, per dollar of debt, under the budget-neutral and net welfare maximizing policies presented in Section 7.2. Panel B reports the mean marginal cost of student debt, defined as the present value of borrowing one more dollar just before graduation.

Read:

- Under current IDR rules, student debt is close to fair value for many undergraduate borrowers, but large graduate balances remain meaningfully subsidized.
- Under SAVE, the average marginal cost of debt drops below 50 cents on the dollar for many undergraduate borrowers, which raises serious moral-hazard concerns.
- Under the paper's optimal policy calibrations, both the average and marginal cost of debt remain close to one, greatly reducing the incentive to overborrow.

5.10 Table 7: Labor-supply response

Read:

- Labor-supply effects are tiny across all policies.
- SAVE slightly lowers labor supply, while the optimal policies slightly raise it.
- The fiscal consequences of labor-supply changes are economically small relative to the much larger direct fiscal consequences of repayment design.

6 Short summary

- Nearly half of the post-2010 rise in student debt reflects **repayment deferral**, not just more borrowing.
- Existing IDR rules create large borrower welfare gains, but a large part of those gains are just **taxpayer transfers**.

Table 7
Fiscal cost from labor supply response.

	Status Quo	SAVE	RAP	Budget-neutral	Highest-welfare
Earnings	42.23	42.23	42.23	42.23	42.23
Taxes and loan repayments	10.64	10.53	10.72	10.97	10.91
Effective marginal tax rate (%)	30.44	30.61	30.38	29.63	30.03
Change in labor supply (%)	–	−0.06	0.02	0.28	0.13
Change tax and loan revenue	–	−0.01	−0.00	0.01	−0.00

This table displays the effect of labor supply responses to changes in IDR rules. Row 1 reports the present value of lifetime earnings. Row 2 reports the present value of taxes and loan payments under alternative repayment rules, holding labor supply fixed. Row 3 shows the income-weighted lifetime effective marginal tax rate, computed numerically by increasing lifetime earnings. Row 4 presents the change in lifetime earnings induced by changes in marginal tax rates, weighted by income. Row 5 reports the resulting change in government tax and loan revenues. Rows 4 and 5 can differ in sign because of covariance terms in the joint distribution of labor supply responses and marginal tax rates. All variables, except in row 3, are in dollar units, and are normalized by the national wage index ($63,795 in 2022).

- The most efficient IDR design is not short forgiveness and heavy subsidization. It is **long repayment horizons with progressive repayment thresholds**, which improve insurance and smoothing at much lower fiscal cost.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that repayment design is quantitatively central for understanding both the rise in student debt and the welfare consequences of student-loan policy.
- It shows that the same IDR program can do two conceptually different things: provide efficient insurance/smoothing and transfer resources from taxpayers to borrowers.
- Once those channels are separated, current US-style IDR rules look much less well designed than they first appear.

7.2 Economic intuition

Student borrowers are young, financially constrained, and exposed to idiosyncratic income risk. They therefore value payment relief most when their income is low or when they are early in the life cycle. Taxpayers, by contrast, value dollars uniformly in present-value terms. This wedge implies that an efficient repayment system should postpone and target payments rather than simply forgive debt early. Long maturities and high repayment thresholds exploit exactly this difference: they save borrowers dollars when those dollars are most valuable, and recover more from them later when the marginal utility cost is much lower.

7.3 Takeaway

The main takeaway is that the policy debate should move away from asking whether student debt should simply be "more forgiven" and toward asking **when**, **for whom**, and **through which margin** repayment should be relaxed. This paper's answer is clear: the high-value use of IDR is to provide insurance and liquidity, not large indiscriminate subsidies. In that sense, the best IDR policy looks less like SAVE and more like a long-horizon, progressive repayment contract that keeps debt close to fair value while still protecting borrowers in bad states and early years.